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Supplementary Material: Agentic Autoresearch for CT Reconstruction

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ABSTRACT

This document contains the supplementary material for *Agentic Autoresearch for CT Reconstruction*. It provides the full solver taxonomy and per-method configuration (S1), the Mayo-LDCT data, split, and geometry pipeline (S2), the breast split and train/test leakage audit (S3), the Mayo test board and significance (S4), the parameter-efficient recombination study (S5), the noise model and no-retrain protocol (S6; the full boards and the framework-axis reading are in the main text), an extended account of agent behavior (S7), and additional results with pointers to the public repository and live leaderboard (S8).

S1 | Solver taxonomy and per-method configuration (Table 1, full)

We place the 26 methods on two axes (Table S1) so that the reversal in the main text can be read off a method's *position* rather than its name. Axis A is *physics engagement*: how much the forward operator **A** is used at inference. It runs from *none* — a purely image-domain map that never revisits the sinogram — through a *single* data-consistency step, to an *in-loop* unrolled scheme that re-applies **A** at every iteration, to a full *per-scene* fit that optimises one network per scan. Axis B is the *prior source* $\mathcal{R}$: where the regularising knowledge comes from. *Hand-crafted* priors (total variation, bilateral filtering) encode smoothness analytically and have no trainable content to overfit. *Supervised* priors are trained on paired low-dose/full-dose examples. *Self-supervised* priors (Noise2Inverse) learn without a clean reference by splitting the measured projections into complementary subsets. *Generative* priors are diffusion or rectified-flow models that synthesise plausible detail. *Implicit/per-scene* priors (neural fields, Gaussian splatting) fit a coordinate network to each individual scan rather than storing a pixel grid. A *foundation* prior is a large network pre-trained once on generic data and applied zero-shot. Each row of Table S1 names a family, its representative solvers, the data-consistency operator **D** each uses, the prior $\mathcal{R}$, the two axis positions, and a source citation; the final row is our own recombination (Section S5), deliberately a hybrid — filtered data-consistency, a learned primal-dual combination, and a hand-crafted bilateral post-filter.

Full per-solver `CONFIG` defaults, parameter counts, and design notes are in each solver's design document under `pentathlon/demo_dl_reference/`, and the complete per-iteration observation records — configuration, reconstruction, and metric for every run — are under `docs/runs/`. The taxonomy is not merely descriptive. Because every solver is grounded in the same differentiable projector, two methods that occupy the same Axis-A/Axis-B cell share the same relationship to the measurements, and — as the reversal shows — they tend to share the same robustness behaviour under noise. That is what lets a method's cell, fixed before any noise is added, predict whether it will prove brittle or robust.

S2 | Mayo-LDCT — data, split, geometry pipeline, and solver implementation

Data and geometry. We use real helical low-dose CT from the 2016 AAPM Low-Dose CT Grand Challenge.[23] The raw acquisition is helical, but our differentiable projector and every solver operate in 2-D fan-beam geometry, so the helical data cannot be

TABLE S1 | Full solver taxonomy. Axis A = physics engagement; Axis B = prior source. The final column cites the source(s) for each family. Abbreviations: DC, data consistency; N2I, Noise2Inverse; rectified-flow prior, a fast diffusion-type generative prior; FoE, field of experts (a learned filter-bank image prior).

Family	Representative solver(s)	Data-consistency D	Prior $\mathcal{R}$	Axis A	Axis B	Ref.
Classical iterative	tv-iterative, manhart-PWLS-TV	filtered / SART grad	total variation	in-loop	hand-crafted	[1,2]
Image-domain denoiser	manduca-bilateral, DD-UNet sup.	none (post-FBP)	bilateral / U-Net	none	hand-cr. / sup.	[3,4]
Unrolled learned-iterative	ITNet v1/v2/v3, U-Swin	in-loop DC	learned CNN / transformer	in-loop	supervised	[5,6,7,8,9]
Variational network	Hammernik-2017, Hammernik-VN	in-loop DC	field-of-experts	in-loop	supervised	[10,11]
Learned primal-dual	learned-primal-dual (LPD)	filtered + learned dual	learned combination	in-loop	supervised	[12]
Dual-domain	DD-supervised, DD-bilateral, DD-N2I	image + sinogram	sup. / bilateral / self-sup.	single/in-loop	sup. / self-sup.	[13,14]
Generative prior	fastdiff flow/wavelet ×4	DC + diffusion	rectified-flow prior	in-loop	generative	[15,16,17,18]
Per-scene implicit	NAF, R^2-Gaussian	per-scene fit	implicit representation	per-scene	implicit	[19,20,21]
Foundation zero-shot	RAM	single DC	frozen foundation model	single	foundation	[22]
Band decomposition	Wu-2015, Wu-2015-trainable	band-weighted DC	frequency-band residual	in-loop	hand-cr. / sup.	[2]
Recombination (ours)	**param-efficient**	**filtered + LPD combine**	**bilateral post-filter**	**in-loop**	**hybrid**	—

used directly. We therefore rebinned helical to fan-beam by single-slice rebinning,[24] which resamples the helical cone-beam rays onto a stack of transaxial fan-beam slices. Getting this right — the detector geometry, the redundancy weighting, and the slice interpolation, so that the low-dose FBP baseline is physically correct rather than merely plausible — was the single hardest engineering task in the study. The differentiable fan-beam projector used at every training and inference step is PYRO-NN,[25] which supplies a matched forward/back-projector pair with analytic gradients, so that the same operator **A** appears in the loss, in every unrolled solver, and in the hr metric.

Effort accounting. We estimate human effort from the version-control history, costing human involvement at 7.5 minutes per human commit — a rate taken from comparable projects, and a deliberately rough gauge, because instructing the agent (a short prompt) and hand-coding (hours of debugging) are very different per-commit costs. We therefore separate the two.

Hand-coded by the human. The helical-to-fan geometry pipeline (`helix2fan`, rebinning, validation scripts) is 33 commits, none of them agent-co-authored — the one component the agent used but did not build. It was debugging-heavy: the effort-timeline record (`docs/runs/mayo_effort_timeline.png`) logs roughly 24 active engineering days for it, far more than 33 commits at 7.5 minutes would imply, which is precisely the point that coding costs more than instructing. This was the dominant human cost of the study.

Agent-coded, human-instructed. The 25 solvers were implemented by the agent in 48 solver-file commits, 41 of them machine-written, with 14 solvers onboarded across two days (Table S2); the three-week benchmark campaign is 443 commits, 439 machine-written, and the agent logged ∼1,100 iterations for ∼134 GPU-hours. The human only instructed here, a few hours of hands-on time at 7.5 minutes per human commit.

Whole project and caveats. About 510 of the ∼1,780 commits are human-authored; at 7.5 minutes each that is ≈64 hours, but this whole-project figure is dominated by results-logging, the dashboard, and the write-up rather than the benchmarking, and it is sensitive to how machine-written provenance commits are classified — so we treat it only as an order-of-magnitude gauge. Finally, the multi-week calendar span overstates the effort: it was stretched by the supervisor's travel and by intermittent, part-time attention, not by continuous full-time work.

Split. We use a fixed patient-level split so that no patient appears in more than one partition: train L145/186/209/219, validation L277, test L014/056/058/075/123. Splitting by patient rather than by slice matters, because adjacent slices of one patient are highly correlated; a slice-level split would leak near-duplicate anatomy between train and test. Test scores are the per-patient mean ± standard deviation over the five test patients ($n = 5$), and this small n is the direct cause of the low statistical power discussed in Section S4. Geometry details and rebinning validation are in the repository. The Mayo challenge is one of a family of AAPM reconstruction grand challenges spanning sparse-view, spectral, truth-based, and metal-artifact tasks.[26,27,28,29,30]

TABLE S2 | Implementation record for the 25 established solvers, read from the git history of the reference implementations. "First" is the first commit touching the solver's file; "#c" is the number of commits that touched it. The four fast-diffusion variants share two files (grouped).

solver	first	#c
itnet	May 15	14
itnet-v2	May 15	13
itnet-v3	May 16	11
uswin	May 16	10
dd-supervised	May 15	14
learned-primal-dual	May 23	3
hammernik-2017	May 16	10
hammernik-vn	May 16	11
tv-iterative	May 15	10
manhart-pwls-tv	May 15	12
manduca-bilateral	Jun 23	1
wu-2015	May 16	10
wu-2015-trainable	May 22	6
dd-n2i	May 15	14
dd-bilateral-sup	May 22	4
dd-bilateral-n2i	May 15	16
tv-iterative-sup	May 23	4
ram	May 18	9
naf	May 16	12
r2gaussian	May 16	12
fastdiff (×4)	Jun 20	3

Implementation effort for the 25 solvers. The solvers themselves were fast to build, and the version-control history of the reference implementations (`pentathlon/demo_dl_reference/solver_*.py`) lets us quantify it. The 25 established solvers were implemented on 22 active days in 63 commits; strikingly, 14 of them were onboarded across only two days (15–16 May 2026, seven per day). The remainder were added in small batches as the study grew: RAM (18 May), Wu-2015-trainable and DD-bilateral-supervised (22 May), learned-primal-dual and TV-iterative-supervised (23 May), the four fast-diffusion variants (20 June), and manduca-bilateral (23 June). Table S2 lists the first-implementation date and the number of commits that touched each solver's file. Two caveats apply when reading time from commits: the agent committed in bursts, so commit *timestamps* bound but do not fully measure the coding-session duration; and several solvers were introduced together in a single commit, so per-solver counts are approximate. The unambiguous signal is nonetheless that turning a published method into a running, benchmarked solver took the agent a fraction of a day, not days — the same speed-up, at the implementation level, that made a 26-method benchmark feasible for one supervisor.

S3 | Breast split and the train/test leakage audit

Split. The Breast CT Challenge releases a public pool of 4000 training cases; its official validation and test truths are withheld for scoring, so we cannot use the challenge's own held-out sets. We therefore re-partitioned the public pool ourselves into train (3600), validation (200), and test (200). The partition is deterministic — a fixed seed (20260703) assigns each case exactly once — and the three subsets are disjoint. Because the truths in our

TABLE S3 | Breast versus Mayo: the two problems are limited by different things, so their absolute hr scales are not comparable.

	Breast CT	**Mayo**
data	noiseless, ideal	real low-dose, quantum noise
bottleneck	data incompleteness	noise / dose
RMSE floor	0	> 0
best hr	∼0.89	∼0.38
dominant lever	data-consistency / known operator	denoising

test split are known to us, the headroom metric can be evaluated directly on it; the price is that we must then prove no test information reaches training. We did so three independent ways.

Leakage audit (three independent checks).

1. *By mechanism.* The training data loader is wired to read only the `train` split; the redirect to the test split fires exclusively inside the evaluation path. No code path exposes a test image during optimisation, so training cannot see test even in principle.
2. *By content.* We hashed every truth image and compared the subsets pixel-for-pixel. Across all 3600 train, 200 validation, and 200 test truths there is zero overlap in every direction — TEST∩TRAIN = 0, TEST∩VAL = 0, VAL∩TRAIN = 0 — so no near-duplicate case straddles the split.
3. *By behaviour.* For every method the test headroom tracks the validation headroom to within about ±0.01, and several methods in fact score *lower* on test than on validation. A model that had memorised the test set would score conspicuously *higher* on test than on validation; the observed pattern is the opposite, which rules memorisation out. The high absolute hr is therefore a property of the data — noiseless and well-posed — and not a leak.

Why breast hr is high, and not comparable to Mayo. The Breast CT Challenge data is noiseless by design; its RMSE floor is zero. Breast is incompleteness-limited (128 of ∼1000 views). Mayo is noise-limited (real quantum noise). The two absolute hr scales are therefore not comparable; Table S3 contrasts the two problems side by side.

S4 | Mayo test board and significance

The board. The champion is ITNet v1 at test hr 0.376, averaged over the five test patients. We test differences with a paired comparison on the per-patient hr values: for each pair of solvers we form the five per-patient hr differences and run a paired t-test, with a Wilcoxon signed-rank test as a distribution-free cross-check. On this basis the top of the board is a statistical tie — ITNet v1, ITNet v2, and U-Swin are indistinguishable at the 5 % level, and the compact param-efficient solver joins this group at the 1 % level ($p = 0.02$–0.027).

Why the tie is unavoidable at $n = 5$. With only five test patients the tests have very little power, and the point is sharpest for the Wilcoxon signed-rank test: with $n = 5$ its smallest attainable two-sided p-value is $2/2^5 = 0.0625$, so it *cannot* reach $p < 0.05$ for any pair, no matter how large the effect. The paired t-test can cross 0.05, but only for the widest gaps. Hence “a tier, not a winner” is not a hedge; it is the honest ceiling of what five patients can resolve, and it is why we report the per-case mean ± standard deviation rather than a bare ranking. The per-patient hr values, the full pairwise significance matrix, and the forest plot are in `docs/runs/mayo_significance_stats.md` and the accompanying figures; the full board (all 26 solvers, hr/SSIM/RMSE) is in the main text.

TABLE S4 | Parameter-efficient recombination arc on Mayo (low-dose, denoising). hr is validation headroom; the last row is the reported low-parameter pick.

step	**architecture**	**params**	**val hr**
CNN prox+DC (seed)	learnable CNN regulariser	∼2800	0.24
FoE filter bank	field-of-experts reg (agent ceiling)	**1921**	**0.25** (ceiling)
+ Manduca projection filter	noise-adaptive sinogram bilateral	1926	0.394
+ multi-scale projection bank	parallel Manduca scales	1937	0.400 (best)
halved FoE (Pareto pick)	n_f 24→12	**969**	**0.360** (test 0.324)

TABLE S5 | Parameter-efficient recombination arc on breast (128-view sparse), all under the 20-min budget.

step	**architecture**	**params**	**val hr**
image-domain unroll (any config)	Mayo-style FoE denoiser	∼1930	≤ 0.26 (ceiling)
filtered DC FBP($\mathbf{Ax} - \mathbf{g}$)	$K = 30$	**13**	**0.4345**
+ learned primal-dual combination	$W = 4$	183	0.5047
+ 3-scale bilateral output filter	$\sigma_r = 0.02$	**195**	**0.6201** (single best)

S5 | The parameter-efficient recombination study (search arc)

The agent recombined the method parts into a minimal-parameter solver on each dataset. Each climb was made of genuine six-box iterations, all under the 20-min budget, and the optimal recombination differs by problem. We present the Mayo arc first, then breast.

Mayo (low-dose, denoising). The seed is a learnable CNN regulariser in a tied prox–DC unroll. Switching the regulariser *family* from a CNN to a field-of-experts (FoE) filter bank set the agent-alone ceiling at hr ≈ 0.25 (Table S4). The break came from the projection domain: prepending a noise-adaptive Manduca bilateral filter to the sinogram, then extending it to a multi-scale bank, lifted val hr to 0.40. Halving the FoE bank yields a 969-parameter Pareto pick (test hr 0.324) — the reported Mayo compact solver.

Breast (128-view sparse). Key steps, all under the 20-min budget:

Findings, each earned by an iteration: the image-domain denoiser caps at ∼0.26; deeper *adjoint* data-consistency lowers hr (it re-injects streaks); the *filtered* gradient breaks the ceiling at 13 parameters; an in-loop prox (proximal / denoising step) hurts (the filtered gradient already reconstructs); a learned cross-step combination helps; a full conjugate-gradient DC solve does not beat the single filtered gradient under a fixed wall; and Wu-2015 band machinery adds nothing here. The champion is seed-fragile in the variational-network manner: 4 of 5 seeds cluster at 0.571 ± 0.033, one collapses; two stabilizers did not recover it. On the held-out test set the compact solver scores hr 0.6212±0.0076 — the tightest per-case std of any solver.

The gains are back-loaded, and human-cued — on both datasets. They are not spread evenly across the ∼40-iteration searches. Under the agent alone the best value plateaued early —

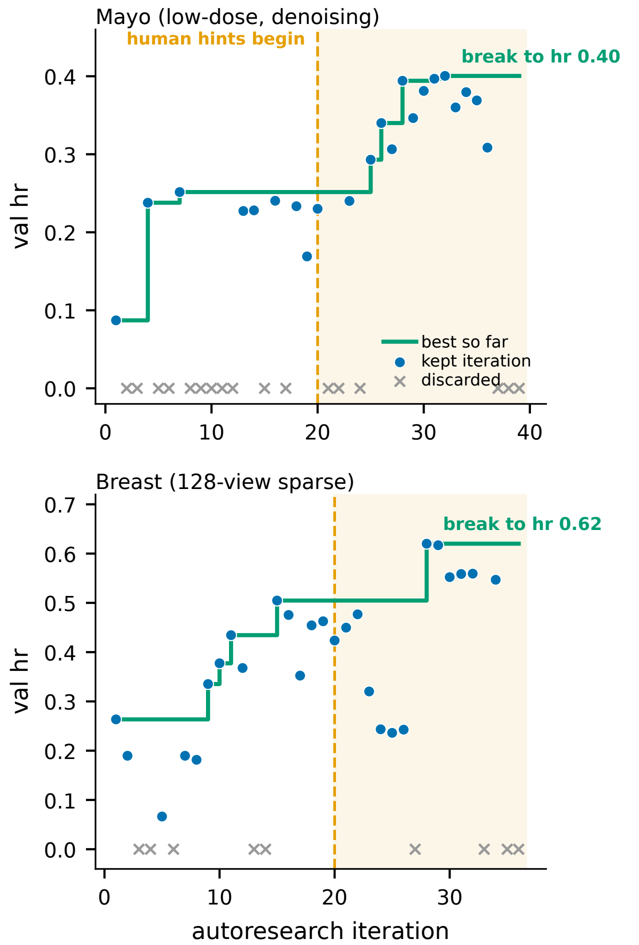

FIGURE S1 | Compact-solver search trajectories. Top: Mayo (low-dose). Bottom: breast (128-view sparse). Points are per-iteration validation headroom (discarded iterations shown at hr 0); the step line is the best value so far. On both datasets the agent's own first 20 iterations plateau (hr ≈ 0.25 Mayo, ≈ 0.50 breast); the decisive break comes only in the hinted second half (shaded) — to hr 0.40 and 0.62 respectively.

at hr ≈ 0.25 on Mayo (by iteration 7) and ≈ 0.50 on breast (by iteration 15) — and then sat flat for a dozen or more iterations. Only in the second half, after a human began suggesting untried directions, did each break through: Mayo to hr 0.40 from iteration 25, breast to hr 0.62 at iteration 28 (Figure S1). This is the clearest evidence in the study of the agent needing human steering to escape a local optimum; without the hints it would have reported both compact solvers at their plateaus.

Mayo vs breast contrast. On Mayo the compact optimum is an evolved field-of-experts plus multi-scale bilateral denoiser (hr 0.324 at $\sim$0.00097 M params), statistically tied with the top tier at the 1 % level. On a denoising problem, a tiny denoiser suffices. On sparse-view breast, that same denoiser fails; data-consistency is the differentiator.[12] This problem-dependence is the practical face of the known-operator principle[31,32]: embedding the exact forward operator cannot raise, and usually lowers, the maximum error bound, and it curbs the hallucination risk of unconstrained learned maps.[33,34] The agent re-derived each optimum; it did not transfer.

Human role in this study. The recombination *strategy* was a human idea, on both datasets. On Mayo the human repeatedly asked whether more parameter-efficient approaches existed and explicitly suggested combining the pieces. On breast the finale was framed as a cross-method recombination and steered away from copying Mayo. The agent executed and mechanistically refined; it did not originate the recombination strategy.

S6 | Noise model and no-retrain protocol

The full noiseless and noisy breast boards, and the framework-axis reading of the reversal, are in the main text; this section records only the exact noise protocol behind the re-evaluation.

Noise model. We convert each clean line-integral sinogram $\mathbf{g}$ to expected photon counts $I_0\, e^{-\mathbf{g}}$, draw Poisson counts, and take the logarithm back to line integrals, following the standard low-dose forward model (Eq. (6) of the main text). The single parameter is the incident photon count I_0; we use $I_0 = 10^5$, a comparatively high dose, so the perturbation is deliberately mild — the RMS of the sinogram perturbation is $\approx$ 0.012 in line-integral units, about 1.8 % at the thickest (most attenuating) ray. The intent is not to stress-test at low dose but to ask whether a *small*, realistic departure from the idealised noiseless data already reorders the leaderboard. It does.

Determinism and fairness. The noise is drawn once per test case from a fixed seed and cached, so every method is evaluated on the byte-identical noisy input. Two solvers that change places do so because of how they process the same measurements, not because they happened to see different noise.

No retraining. Supervised solvers load the checkpoint they trained on the noiseless data and run inference only — their weights never see a noisy example. Classical and per-scene solvers (TV, PWLS-TV, bilateral, NAF, R^2-Gaussian) carry no pre-trained weights and simply re-run their normal inference-time fit on the noisy input, which is their intended mode of use. The FBP baseline in the headroom metric is recomputed from the *noisy* sinogram, so hr measures each method's improvement over the noisy FBP and every method on the board is scored against the same baseline. The comparison is thus fair *within* the noisy board, even though the noisy and noiseless hr scales are not directly comparable.

S7 | Agent-behavior — extended account

This account follows the autoresearch paradigm of automated open-ended scientific discovery[35] realised through LLM code agents that resolve real software tasks[36,37] and physics-informed agentic development in adjacent imaging fields.[38] We summarise here what the agent did well, where it failed, and which parts of the study still required a human.

Strengths. The agent is fast from paper to code: it read a method description and produced a running solver in a fraction of a day (Section S2), which is what made 26 methods across two datasets tractable for one supervisor. It is a strong hyper-parameter optimiser, moving one knob at a time against the frozen metric and reliably driving each solver to its ceiling. It keeps a fixed compute budget without drift — the 20-minute per-iteration wall was honoured across every run, a discipline humans find tedious. It scales evaluation almost for free, launching and tracking dozens of independent benchmark jobs in parallel; this parallelism, not raw single-task speed, is its most valuable practical property. And it follows well-decomposed instructions faithfully, provided each sub-task is stated concretely.

Weaknesses. The agent does not invent new methods; it implements, tunes, and recombines existing ones, and it proposed no new reconstruction principle. It must be *forced* to recombine: left unsupervised it converges early and then pads — in one breast sub-run it replayed near-identical configurations until an audit caught the repetition. Its CT-image “vision” is unreliable: it repeatedly failed to see obvious streaks and artefacts in reconstructions and sinograms, so throughout the study the numeric metric, never the rendered image, had to be the source of truth. Long tasks must be decomposed into sub-tasks; without decomposition even a one-million-token context is exhausted before the task finishes. Finally, it overfits to the task and the chosen metric — though, as the reversal shows, that failing is not unique to agents; human researchers overfit to benchmarks too.

Human meta-strategies (repeatable, and the same on both datasets). Four human interventions recurred and mattered. *Persistence and breadth*: repeatedly asking the agent for more directions once it had stalled. *Recombination*: supplying the idea of assembling proven pieces into one compact solver — the decisive move behind the parameter-efficient result on both datasets (Section S5). *Redirection*: forbidding the agent from copying the other dataset’s answer, which forced it to re-derive each optimum from local evidence. *Auditing*: catching padding, enforcing the compute budget, and repairing provenance. The defensible division of labour is therefore this: the agent is a tireless, mechanistically reasoning, self-modifying executor, and the human is the strategist and the auditor.

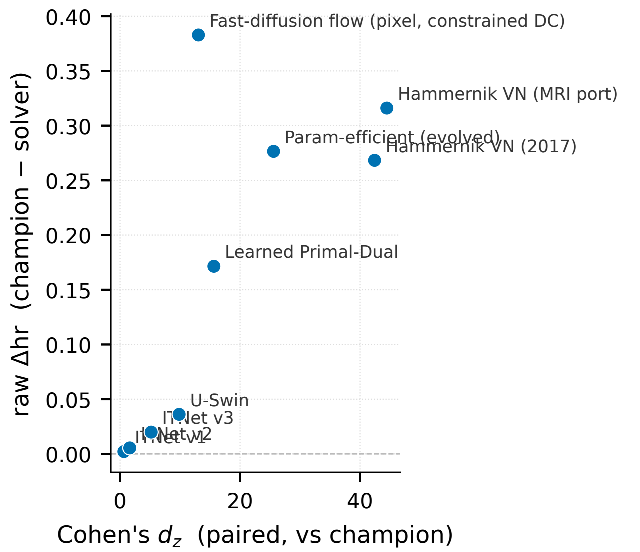

FIGURE S2 | Effect size on the breast board ($n = 200$): Cohen’s d_z versus raw Δhr against the champion, one point per solver. Large $|d_z|$ at $n = 200$ is why the breast board separates cleanly, unlike the $n = 5$ Mayo board.

S8 | Additional results and the live leaderboard

One further figure is included here. Figure S2 shows, on the breast board ($n = 200$), the effect size (Cohen’s d_z) against the raw Δhr to the champion for each solver; with $n = 200$ even small hr gaps are highly significant, in contrast to the low-power $n = 5$ Mayo setting. The parameter/accuracy trade-off itself — for both datasets — is Figure 2 of the main text, and the per-step climb is quantified in the recombination-arc tables above (Tables S4 and S5).

Everything that did not fit the paper is public. The complete results live in the repository and the live dashboard:

https://github.com/akmaier/Agent4CT
https://akmaier.github.io/Agent4CT/

The dashboard hosts the full leaderboards for all datasets — Mayo, noiseless breast, and the noisy re-evaluation — with every solver’s per-iteration trajectory. Each iteration carries an immutable provenance record: the exact configuration, the reconstruction, and the frozen metric, so any number in this paper can be traced to the run that produced it. The repository additionally holds the Mayo significance forest plot and solver×metric matrix, the val-vs-test selection-honesty curves, the effort-timeline figure, the noiseless-versus-noisy reconstruction montages for the top solvers, and the per-solver design docs and `CONFIG` defaults.

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